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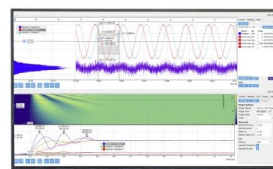
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A finite element approach to forward modeling of nuclear magnetic resonance measurements in coupled pore systems

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ABSTRACT

Porous media characterized by a hierarchy of length scales are ubiquitous in industry and nature, and include carbonate rocks, cements, heterogeneous catalysts, and biological cells. Nuclear magnetic resonance (NMR) is a popular tool for studying liquid-saturated porous materials, where the spin relaxation rate is generally considered proportional to pore size. However, in porous granular media, the relaxation rate is modified by diffusion between the intraparticle and interparticle pores. The observed relaxation rates do not reflect the pore size under such conditions. Deconvolving the various contributions of surface relaxation, geometry, and diffusion is nontrivial, and forward models are a powerful technique for elucidating the underlying pore structure. Various forward models have been proposed previously, including analytic solutions and random walk simulations. Here, a finite element method is adopted to simulate the diffusion of nuclear magnetization in a coupled pore geometry. We validate our model against existing solutions and use the simulations to determine the surface relaxivity of powdered silica by matching experimental results. The finite element approach is more versatile than other modeling methods, allowing direct visualization of the diffusing magnetization and being trivially extensible to multidimensional NMR exchange experiments.

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I. INTRODUCTION

Nuclear magnetic resonance (NMR) relaxation time measurements are commonly used to probe the structure of liquid-saturated porous media. Information on the confining geometric length scales is recovered by assuming that the observed relaxation rate (inverse of time) is proportional to the surface-to-volume ratio S/V of the pores. In complex materials, characterized by a hierarchy of length scales, distributions of longitudinal T_1 or transverse T_2 relaxation time are used as a proxy for pore size distributions.¹ This interpretation is based on the assumption that each pore is isolated from its neighbors.² However, if pores of substantially different sizes are strongly coupled, this interpretation is no longer valid.³

Materials characterized by coupled pore systems include limestone,^{3–5} heterogeneous catalysts,^{5,7} and cements.^{8–10} We refer to these materials as porous granular media as they comprise

connected “micropores” inside grains separated by “macropores” (interparticle voids, vugs, fractures, or cracks). NMR has been applied ubiquitously to the study of these heterogeneous materials despite the persistent challenge of deconvolving the contributions of surface relaxation and diffusion between the micropores and macropores. Multidimensional NMR relaxation time correlations, such as T_1 - T_2 ¹¹ and T_2 - T_2 ,¹² are useful to elucidate the structure-transport relations in analogy to chemical exchange spectroscopy.¹³ While these experiments highlight the presence of exchange^{14,15} and can provide estimates of diffusional coupling constants,^{5,9,16,17} the challenge of extracting geometric length scales persists.

Forward modeling has been used to elucidate the underlying pore structure. A model based on prior knowledge of the sample (porosity, grain size, liquid diffusion coefficient) is tested against experimental NMR data to estimate the unknown parameters (pore size, surface relaxivity, diffusional coupling constant).

Several techniques have been applied to this problem. Analytic solutions range from simple two-site exchange models^{9,17,18} to a one-dimensional (1D) porous grain.^{3,4,19} Random walk simulations in a three-dimensional (3D) periodic lattice (microporous grain consolidation model, μ GC) have been used extensively to support the interpretation of NMR relaxation time measurements.^{3,20–23} Both the analytic and random walk approaches have limitations. The analytic two-site exchange model is used to describe multidimensional NMR experiments^{9,17,22,24} but assumes spatially uniform magnetization. The analytic 1D porous grain model has not been extended to represent multidimensional NMR experiments, and an approximate extension to three spatial dimensions (based on the μ GC lattice) assumed uniform magnetization in the macropores.³ The random walk simulations are computationally expensive, and even though each walker is independent giving rise to readily parallelizable code, simulations of a T_2 - T_2 correlation can run to days on a desktop personal computer (PC).²¹ Furthermore, our available random walk simulation code is based on a primitive cubic lattice of spherical solids and cannot readily accommodate arbitrary grain or pore geometries. However, more sophisticated random walk methods have been presented in the literature based on X-ray microcomputed tomography (μ -CT) images with embedded microporosity.^{25,26}

Here, we adopt a finite element method to model the diffusion of magnetization in a coupled pore system. For convenience, we employ the commercial COMSOL Multiphysics[®] package. We have previously used this software to explore T_2 relaxation in the presence of pore-scale magnetic field inhomogeneities²⁷ and simulate two-site exchange.²⁸ To simulate the diffusion of magnetization in a granular porous medium, the model geometry is divided into two continuum domains: the intergrain macropore and the microporous grain.¹⁹ Due to the orders of magnitude difference in length scale between the individual micropores and the macropore, it is not practical to generate and mesh the exact geometry. Therefore, the interior of the grain is represented as a spatially uniform domain with volume-averaged properties (grain porosity ϕ_μ , apparent diffusion coefficient $\mathcal{D}_{app}$, and relaxation time T_μ); these spatially averaged properties are calculated in a separate model representing a single micropore. Coupling between the microporous grain and the macropore is established by way of matched boundary conditions on either side of a smooth interface. All outer boundaries are reflecting to represent a periodic geometry.

The general theory describing the diffusion of continuum nuclear magnetization in a porous granular medium is presented in Sec. II. An example periodic geometry is illustrated in Sec. III based on the μ GC model of Schwartz,²⁹ and we introduce the various simulations based on this geometry. The numerical methods used to estimate relaxation time distributions from the magnetization decay data are outlined in Sec. IV. The results obtained by finite element analysis of a 1D geometry and analytic solutions are compared in Sec. V A, and results from finite element analysis of a 3D μ GC geometry are compared to random walk simulations in Sec. V B. These comparisons serve to validate the use of COMSOL Multiphysics for simulations of magnetization diffusing in a coupled porous medium. We then extend the finite element method to multidimensional NMR experiments and generate T_1 - T_2 correlations from the μ GC geometry. Finally, we compare experimental T_1 - T_2 correlations to the simulations performed under

matching conditions in Sec. VII and show that the salient features of the experimental data are captured by the coupled pore model.

II. THEORY

A. Magnetization in a single pore

We first consider the case of an isolated pore. The spatial and temporal dependence of the nuclear magnetization $m(\mathbf{r}, t)$ is governed by the Bloch-Torrey equation³⁰

$$\frac{\partial m(\mathbf{r}, t)}{\partial t} = \mathcal{D}_0 \nabla^2 m(\mathbf{r}, t) - \frac{m(\mathbf{r}, t)}{T_b}, \quad (1)$$

where $\mathcal{D}_0$ is the bulk diffusion coefficient and T_b is a bulk relaxation time. The partially absorbing boundary condition

$$\mathcal{D}_0 \hat{\mathbf{n}} \cdot \nabla m(\mathbf{r}, t) + \rho m(\mathbf{r}, t) = 0 \quad (2)$$

applies at the pore wall, where $\hat{\mathbf{n}}$ is the surface-normal vector and ρ is the surface relaxivity. Brownstein and Tarr provided a solution to these equations using an eigenfunction-eigenvalue expansion.² For diffusion in a 1D slab of width $2L$ under conditions of “fast diffusion,” such that the dimensionless parameter $\alpha = \rho L / \mathcal{D}_0 \ll 1$, a single eigenfunction-eigenvalue pair is sufficient to describe the decay of nuclear magnetization, and the familiar relationship

$$\frac{1}{T} = \frac{\rho}{L} \equiv \rho \frac{S}{V} \quad (3)$$

applies exactly. Equation (3) may be applied to transverse relaxation with T_2 and ρ_2 or longitudinal relaxation with T_1 and ρ_1 . If the surface relaxivity is known, the measured relaxation time can be rescaled to pore size. Equation (3) is approximate (within a factor unity²) for a 2D cylinder of radius $r = 2L$ or a 3D sphere of radius $r = 3L$. If a hierarchy of length scales exists within the porous medium, then it is reasonable to expect that multiple relaxation times will be observed. Assuming the pores are effectively isolated, such that individual spins explore only a single geometric length L (i.e., $V/S < \sqrt{\mathcal{D}_0 t}$, where $t \sim T_1, T_2$ is the experimental time scale)³¹ and $\alpha \ll 1$, then a distribution of relaxation times is equivalent to the pore size distribution.¹ However, if the diffusing spins explore multiple length scales during the experiment (or $\alpha \gg 1$), the measured relaxation times will not reflect the pore size.

B. Magnetization in coupled pores

Now consider the case of a porous granular material. The system is characterized by two length scales: an intragrain micropore size $L_\mu \sim 1 \mu\text{m}$ and an intergrain macropore size $L_m \sim 100 \mu\text{m}$. Note the ratio $L_m/L_\mu \sim 100$. The pore network is saturated with a single liquid of bulk diffusivity $\mathcal{D}_0 \sim 10^{-9} \text{m}^2 \text{s}^{-1}$. For realistic values of $\rho < 10^3 \mu\text{m s}^{-1}$, it is reasonable to assume $\alpha < 1$ in the micropores, so the relaxation is characterized by a single time T_μ (local magnetization m_μ is spatially uniform inside a pore) and included in the diffusion equation (1) as an additional constant rate term. Accordingly, the microporous grain may be represented as an effective medium with the diffusion of magnetization governed by

$$\frac{\partial m_\mu}{\partial t} = \frac{\mathcal{D}_0}{\kappa} \nabla^2 m_\mu - m_\mu \left(\frac{1}{T_\mu} + \frac{1}{T_b} \right), \quad (4)$$

where κ is the tortuosity of the pore network, defined as

$$\lim_{t \rightarrow \infty} \left[\frac{D_{\text{app}}}{D_0} \right] = \frac{1}{\kappa} \quad (5)$$

for local grain porosity ϕ_μ and apparent diffusion coefficient D_{app} .^{32,33} For simplicity, we ignore the time-dependence of the diffusion coefficient in the microporosity and take the long-time limit as an appropriate approximation. In petrophysics, the tortuosity of a porous rock is expressed in terms of the formation factor $F = \phi^{-c_{\text{cem}}} \equiv \kappa/\phi$,³⁴ where the cementation exponent c_{cem} is determined empirically. For uniform microporous mudstones, c_{cem} will typically take values close to 2. For synthetic porous media as studied in Sec. VII, we take $c_{\text{cem}} = 1.5$, so the tortuosity is $1/\kappa = \sqrt{\phi}$ after Davies *et al.*³⁵

The diffusion of magnetization m_m in the macropores is unrestricted and is therefore governed by Eq. (1). At the interface between the microporous grains and the macropores, the magnetization must be continuous, giving the simple continuity condition

$$m_m = m_\mu. \quad (6)$$

The flux of magnetization must also be matched across the interface according to³

$$D_0 \hat{n} \cdot \nabla m_m + \rho m_m (1 - \phi_\mu) = \frac{\phi_\mu D_0}{\kappa} \hat{n} \cdot \nabla m_\mu. \quad (7)$$

The total porosity of the granular medium is

$$\phi = f_m \phi_m + f_g \phi_\mu, \quad (8)$$

where f_m is the fraction of the sample volume occupied by the macropore, the local porosity in the macropore $\phi_m = 1$ by definition, and f_g is the fraction of the sample volume occupied by grain.

III. GRAIN CONSOLIDATION MODEL

The μ GC model²⁹ is a simple periodic lattice of microporous spherical grains, as illustrated in Fig. 1. Each grain is formed of spherical nonporous particles uniformly distributed on a regular sublattice. At the lattice or sublattice length scale, spherical particles of radius R are arranged on a simple grid [Fig. 1(a)] to define the cubic unit cell of side length A shown in Fig. 1(b). At the lattice scale, the porous grains have radius R_m and the unit cell has a side length A_m ; at the sublattice scale, the solid particles have radius R_μ and the unit cell has a side length A_μ . The ratio $A_m/A_\mu \sim 100$.

A. Random walk simulations

Random walk simulations were performed directly in the μ GC geometry using a variable time step (first arrival) technique.^{36,37} Walkers (nuclear spins) were seeded over the entire unit cell, and those assigned to the solid volume were rejected. The ratio of accepted to rejected walkers provided an estimate of the porosity. Walkers diffused to a solid surface with an arrival time determined by an analytic probability function. Once a walker was within a distance 3ϵ of a solid surface, its motion was governed by fixed jumps of length $\epsilon = 0.01 \mu\text{m}$. When in proximity to the surface, the walker had a probability γ of relaxing, determined by the surface relaxivity

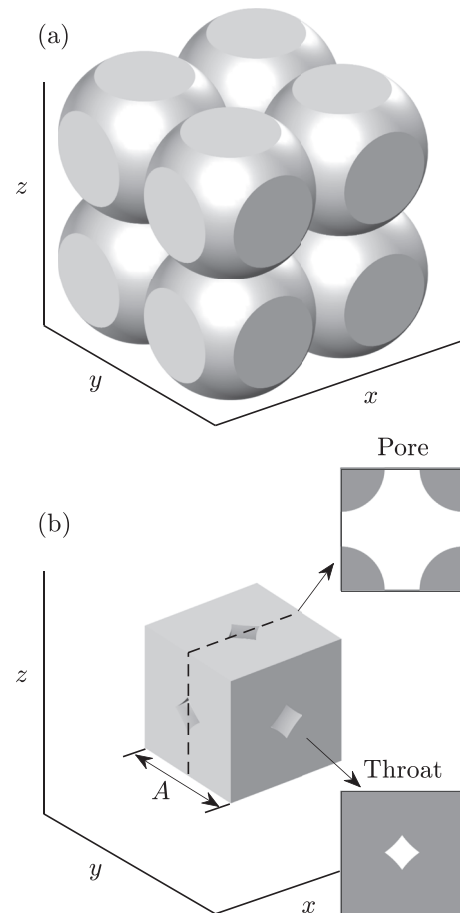


FIG. 1. Grain consolidation model after Ramakrishnan *et al.*³ Spherical particles of radius R are arranged on a simple grid (a) to define a unit cell (b) of side length A . The illustrated spheres in (a) are oversized such that $2R/A = 1.23$. The insets in (b) show cross sections of the unit cell corresponding to an enclosed pore and a connecting throat. The same geometric structure is used at the lattice (macropore) and sublattice (micropore) scales.

such that $\gamma = (2\epsilon\rho)/(3D_0)$.³⁸ If a walker relaxed, it was deleted. At fixed time intervals, the number of surviving walkers was counted to determine the fraction of observable magnetization. A total of 100 000 walkers were run in each simulation (not including discarded walkers initially assigned to the solid volume), and this number of walkers was sufficient to produce consistent T_2 distributions on repeat simulations. Note that the random walk simulations captured the full detail of the structure at all length scales and there was no requirement to represent the microporous grains as a continuous medium.

B. Analytic solution in 1D

The μ GC model is defined explicitly in three dimensions but is approximated in 1D by the geometry in Fig. 2.³ Each unit cell of the 1D model repeats periodically over a length $2W$, although it is only necessary to simulate the symmetric geometry over $0 < x < W$.

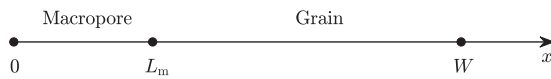


FIG. 2. One-dimensional porous grain geometry used for analytic calculations and finite element simulations of diffusing magnetization. The boundaries at $x = 0$ and W are reflecting. The mixed pore/grain boundary at $x = L_m$ is absorbing and porous.

The macropore domain has length L_m , and the grain domain (represented by an effective continuous medium) has length L_μ so that $W = L_m + L_\mu$ with the pore/grain interface at $x = L_m$. The boundaries at $x = 0$ and W are reflecting.

An analytic solution to Eqs. (1) and (4) given the interfacial boundary conditions (6) and (7) is presented elsewhere.^{3,4} The equations were reduced to one spatial dimension and solved for $m(x, t)$. A solution was obtained via the method of Laplace transforms as explained previously.³⁹ The time-dependent magnetization was transformed to

$$\tilde{m}(s) \equiv \int_0^\infty m(t) \exp\{-st\} dt, \quad (9)$$

where the residuals of the poles $\{s_n\}$ of $\tilde{m}(s)$ in the complex s -plane, calculated from

$$m(t) = \sum_{n=0}^\infty \lim_{s \rightarrow s_n} (s - s_n) \tilde{m}(s_n) \exp\{s_n t\}, \quad (10)$$

are the discrete relaxation rates $1/T_n = -s_n$, each with a corresponding amplitude and mode shape.

For magnetization in the microporous grain domain $x > L_m$, it was assumed $\alpha < 1$, so the relaxation behavior was approximated by Eq. (3) using a defined value of ρ and the solid particle surface-to-volume ratio S_μ/V_μ . For magnetization in the macropore domain $x \leq L_m$, the partially absorbing grain surface had an apparent relaxivity³

$$\rho_{app} \approx \rho(1 - \phi_\mu) + \phi_\mu \sqrt{\frac{D_0}{\kappa T_\mu}}, \quad (11)$$

which led to an estimate of the dominant macropore relaxation rate

$$\frac{1}{T_m} \simeq \frac{\rho_{app}}{L_m}. \quad (12)$$

For modest values of ϕ_μ , κ , and D_0 , it is possible for $\rho_{app} \gg \rho$. To represent the 3D interstices in the μ GC model as 1D slabs, the geometric lengths were taken as $L_m \equiv V_m/S_m$ and $U \equiv V_\mu/S_\mu$ (where U is the effective size of an individual micropore such that the 1D grain domain represents an ensemble average over $L_\mu/U \sim 200$ micropores), and the diffusion coefficient reduced by an appropriate factor. The micropore relaxation time was calculated from Eq. (3) as $T_\mu = U/\rho$. Estimating V/S for the interstice of oversized spheres is nontrivial, although approximate analytic integrals exist. Qualitatively consistent results have been generated previously using the 1D analytic solution and 3D random walk simulation with matched parameters.³

The efficiency of the coupling between the microporous grain and the macropore is represented by a dimensionless parameter C ,¹⁹ defined in our notation as

They are quoting Annand2007 on this, but the equation is not quite the same as in Annand. here L_m is the "macropore domain" while for Annand is the size of the microporous grain.

$$C = \frac{\phi_\mu(1 - \phi_m)}{\phi} \frac{\rho L_m^2}{D_0 U}. \quad (13)$$

This coupling constant is equivalent to the Brownstein-Tarr parameter inasmuch as $C \gg 1$ corresponds to strong surface relaxivity and a magnetization decay that is insensitive to diffusive exchange between the grain and macropore domains.

C. Finite element simulations

The μ GC model was constructed using the COMSOL Multiphysics drawing tools in 3D (Fig. 1) or 1D (Fig. 2). The model was divided into two domains corresponding to the macropore and the microporous grains, and the grains were treated as an effective continuous medium. A single unit cell was constructed for each simulation. The Bloch-Torrey equations governing the diffusion of magnetization in the macropore domain Eq. (1) and the grain domain Eq. (4) were established using general form partial differential equations. The boundary conditions were set at the interface between domains according to Eqs. (6) and (7). A time-dependent study was used to solve for $m(\mathbf{r}, t)$ across both domains simultaneously.

It was necessary to determine the relaxation behavior in the microporous grains for both the 3D and 1D finite element simulations. To obtain the input parameters for the continuum porous grain domain, we performed a separate calculation in a single micropore, as illustrated in Fig. 3. Diffusion of magnetization inside the pore proceeded as bulk diffusion according to Eq. (1) with the absorbing boundary condition Eq. (2) at the solid walls. The other pore surfaces, corresponding to the throats indicated in Fig. 1(b) (inset), were reflecting only. The input parameters A_μ , R_μ , and ρ were used to calculate ϕ_μ , T_μ , and U ; the relaxation time was determined by fitting a single exponential function to $M_\mu(t)$, as defined in Sec. IV below. These derived quantities were used as inputs to the main

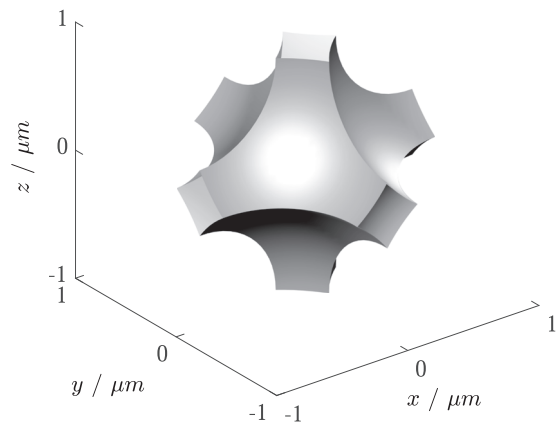


FIG. 3. Geometry used to calculate ϕ_μ and T_μ in micropores with input parameters A_μ , R_μ , and ρ or f_m and T_m in macropores with input parameters A_m , R_m , and ρ_{app} . This geometry corresponds to the interstice of Fig. 1(b). Diffusion of magnetization occurs inside the pore with an absorbing boundary condition at the walls. The outermost (flat) surfaces correspond to connecting throats and are reflecting only. The example micropore geometry shown was generated with $A_\mu = 2 \mu\text{m}$ and $2R_\mu/A_\mu = 1.23$.

simulation. The tortuosity was taken as $\kappa = 1/\sqrt{\phi_\mu}$ from Eq. (5). For the 1D simulation, it was also necessary to specify L_m . The single pore (interstice) geometry in Fig. 3, rescaled to A_g and R_g , was used to calculate $L_m \equiv V_g/S_g$.

IV. DATA PROCESSING

All the simulation methods generate data that describe the exponential decay of magnetization in the time domain. The signal is recorded as an average of the magnetization over the entire geometry

$$M(\tau) = \frac{\int m(\mathbf{r}, \tau) dV(\mathbf{r})}{\int dV(\mathbf{r})} \quad (14)$$

to give the normalized amplitude as a function of time τ according to the first-kind Fredholm integral equation

$$\frac{M(\tau)}{M(0)} = \int_0^\infty f(T) \exp\left\{-\frac{\tau}{T}\right\} d(\log_{10} T) + \epsilon(\tau), \quad (15)$$

where $f(T)$ is the required distribution and $\epsilon(\tau)$ represents noise. In most cases, $m(\mathbf{r}, 0) = 1$ everywhere, so $M(0) = 1$ making the normalization redundant. The recorded data are essentially noise-free except in the case of the random walk simulations where the finite number of walkers introduces noise-like fluctuations on the output. We make no distinction between longitudinal and transverse magnetization at this stage and keep the notation general. Both relaxation processes may be represented as exponential decays without consequence for the interpretation, and the only distinguishing parameter that needs to be considered in simulation is the choice of surface relaxivity.

Distributions of relaxation time $\hat{f}(T)$ are estimated on the $\log_{10} T$ scale by numerical inversion of the data described by Eq. (15). This inverse problem is ill-posed, so we use Tikhonov regularization to determine a stable solution over a limited range of T with a second-difference smoothing operator and smoothing scalar selected by generalized cross validation (GCV) as described by Wilson.⁴⁰

Experimentally, transverse magnetization is observed to decay in the time domain, so Eq. (15) is directly applicable to measurements of T_2 by the conventional Carr-Purcell-Meiboom-Gill (CPMG) spin echo method,^{41,42} where the echo amplitudes (magnetization) are recorded at constant time intervals t_e and the acquisition time of the n th echo is $\tau_2 = nt_e$. Longitudinal relaxation may also be encoded in an exponential decay function⁴³ although it is more common to determine T_1 by observing the recovery of magnetization to its equilibrium state.⁴⁴ The NMR signal amplitude (magnetization) is recorded at time τ_1 which may be chosen arbitrarily and is incremented in successive acquisitions to explore the time-domain behavior.

Multidimensional NMR experiments are constructed by sequential application of methods, such as observing longitudinal magnetization recovery followed by transverse magnetization decay in a T_1 - T_2 correlation experiment.¹¹ Data are only recorded over the second experimental section, namely, the transverse magnetization decay encoded by CPMG. To generate equivalent data in simulation, two separate COMSOL Multiphysics simulations were performed. For simplicity, both longitudinal and transverse relaxation processes

were encoded as exponential decays. In the first simulation, the (longitudinal) magnetization was modeled in the porous system with surface relaxivity ρ_1 . A set of solutions $m(\mathbf{r}, \tau_1)$ was calculated over the required range of τ_1 . Each solution was then used as the input for the second simulation of (transverse) magnetization and the signal (spatial average) stored at intervals of t_e to span the required range of τ_2 . These data are represented by the two-dimensional Fredholm integral equation

$$\frac{M(\tau_1, \tau_2)}{M(0, 0)} = \int_0^\infty f(T_1, T_2) k_1(\tau_1, T_1) k_2(\tau_2, T_2) \times d(\log_{10} T_1) d(\log_{10} T_2) + \epsilon(\tau_1, \tau_2), \quad (16)$$

where the kernel functions have the form

$$k_i(\tau_i, T_i) = \exp\left\{-\frac{\tau_i}{T_i}\right\}, \quad i = 1, 2. \quad (17)$$

The initial magnetization is again defined such that $M(0, 0) = 1$. When $T_1 \neq T_2$, the transfer of magnetization between the longitudinal and transverse eigenfunctions results in contributions to $f(T_1, T_2)$ with negative amplitude and $T_2 > T_1$. The popular approach to inverting data described by Eq. (16) introduces a non-negativity constraint on the solution as part of the regularization step^{45,46} and distorts the resultant distribution when negative-amplitude components are present. Consequently, we use a primal-dual algorithm to solve the optimization problem without constraining the output sign, following the method of Chandrasekera and Mitchell;²⁸ all the T_1 - T_2 correlation plots presented here were generated using this inversion method.

Experimental T_1 - T_2 correlations were measured on powdered microporous silica saturated with de-ionized water. The silica powders were manufactured by Fisher Scientific with quoted particle size ranges of $R_m = 3$ –17.5 μm , 20–31.5 μm , 30–100 μm , and 100–250 μm . The small particle size (3–17.5 μm) had a quoted micropore size of 4 nm, and all the other silica powders had a quoted micropore size of 6 nm. The measurements were performed on an Oxford Instruments bench top magnet ($B_0 = 0.3$ T corresponding to $f_0 = 12.9$ MHz for ^1H). The longitudinal relaxation was encoded using inversion recovery⁴⁴ with the recovery delays incremented logarithmically from $\tau_1 = 1$ ms to 10 s in 32 separate acquisitions. CPMG echo trains were recorded with $t_e = 400$ μs and 5000 echoes. The integral equation (16) describing the experimental data was solved using the kernel functions

$$k_1(\tau_1, T_1) = 1 - 2 \exp\left\{-\frac{\tau_1}{T_1}\right\}, \quad (18)$$

$$k_2(\tau_2, T_2) = \exp\left\{-\frac{\tau_2}{T_2}\right\}. \quad (19)$$

As noted above, the estimated two-dimensional distribution $\hat{f}(T_1, T_2)$ is independent of the encoding method.

V. VALIDATION OF FINITE ELEMENT SIMULATIONS

In this section, we compare simulated results obtained using the COMSOL Multiphysics package to outputs from the existing analytic solution (1D) and random walk simulations (3D).

A. Comparison of finite element simulations to analytic solutions in 1D

The 1D geometry in Fig. 2 was constructed using parameters selected to approximate the 3D μ GC model presented by Ramakrishnan *et al.*³ The input parameters were $W = 38.34 \mu\text{m}$, $L_m = 6.49 \mu\text{m}$, $\phi_\mu = 0.1694$, $U = 0.16 \mu\text{m}$, $\kappa = 2.43$, $T_b = \infty$, and $D_0 = 1.0 \times 10^{-9} \text{m}^2 \text{s}^{-1}$. The magnetization decay was calculated by finite element simulation and analytic solution for surface relaxivities ranging from $\rho = 1.5 \mu\text{m s}^{-1}$ to $937.5 \mu\text{m s}^{-1}$ (multiples of 5), where the two lowest values of ρ were considered representative of real porous materials,³ although experimentally determined ρ values depend on the method used to measure S/V .⁴⁷ The calculated micropore and dominant macropore relaxation times are listed in Table I. The relaxation time distributions are shown in Fig. 4 (left). The magnetization decays generated by simulation were converted to pseudocontinuous distributions by numerical inversion, whereas the analytic solution provided discrete relaxation times and amplitudes. A large number of poles (and hence relaxation rates) were calculated using the analytic solution, but most were associated with negligible amplitude. Only the components of significant amplitude are shown in Fig. 4 for clarity. The distributions are bi-modal except for the highest surface relaxivity [Fig. 4(i)]. Excellent agreement is observed in the results from the two methods, as expected given that the underlying physics is identical. The peaks in the distributions from simulation appear lower than the discrete analytic component amplitudes because of the smoothing imposed by the numerical inversion. However, it is clear from the cumulative integrals in Fig. 4 (right) that the two sets of distributions are consistent. These results validate the solutions provided by finite element simulation.

Indicated in Fig. 4 (left) are the calculated micropore and dominant macropore relaxation times, scaled to effective pore size. In all cases, the fastest relaxation time in each distribution corresponds almost exactly to T_μ . When the surface relaxivity is weak, e.g., $\rho = 1.5 \mu\text{m s}^{-1}$ or $7.5 \mu\text{m s}^{-1}$, see Figs. 4(a) and 4(c), respectively, then the slowest relaxation time corresponds to T_m , albeit with the effective pore size much smaller than the true pore length $\rho T_m \ll L_m$. However, as the surface relaxivity becomes stronger through Figs. 4(e), 4(g), and 4(i), the slowest relaxation time increases above T_m . It is interesting to note that only in Fig. 4(g) does the effective macropore size correspond to the true pore length $\rho T_m \approx L_m$. The observed shift in relaxation time corresponds to a

TABLE I. Variation in the predicted micropore and macropore relaxation times as a function of surface relaxivity and the corresponding pore size based on Eq. (3). Also included are the Brownstein-Tarr parameter $\alpha_m = \rho L_m / D_0$ for the macropore only and the grain-macropore coupling parameter C as defined in Eq. (13).

ρ ($\mu\text{m s}^{-1}$)	T_μ (ms)	ρT_μ (μm)	T_m (ms)	ρT_m (μm)	α_m	C
1.5	108.3	0.16	602.4	0.90	0.01	0.18
7.5	21.7	0.16	226.5	1.70	0.05	0.90
37.5	4.3	0.16	78.8	2.95	0.24	4.48
187.5	0.9	0.16	23.9	4.48	1.23	22.41
937.5	0.2	0.16	6.3	5.86	6.08	112.03

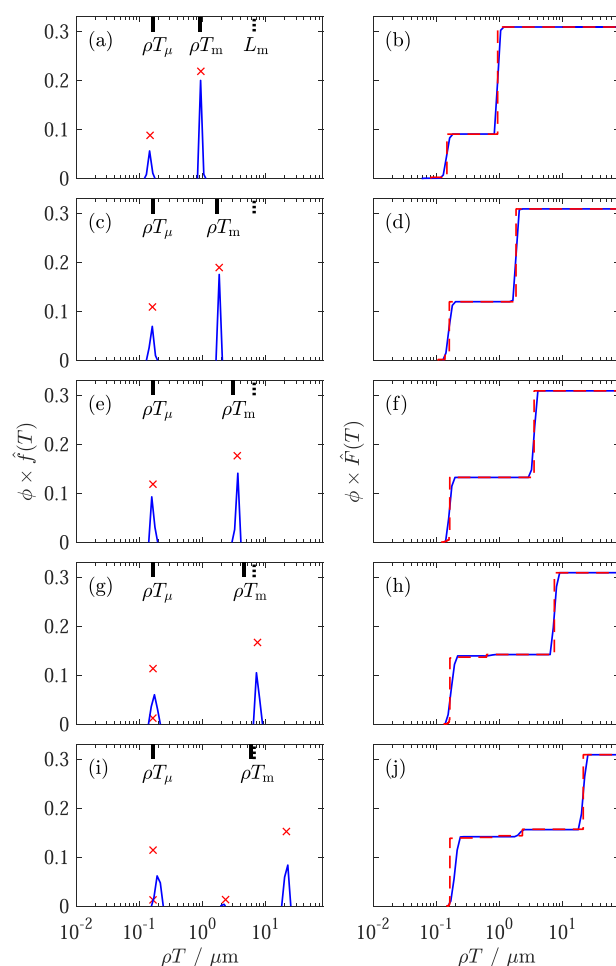


FIG. 4. Simulated relaxation behavior in a 1D μ GC analog geometry. The surface relaxivity varied as [(a) and (b)] $\rho = 1.5 \mu\text{m s}^{-1}$, [(c) and (d)] $7.5 \mu\text{m s}^{-1}$, [(e) and (f)] $37.5 \mu\text{m s}^{-1}$, [(g) and (h)] $187.5 \mu\text{m s}^{-1}$, and [(i) and (j)] $937.5 \mu\text{m s}^{-1}$. See text for other simulation parameters. The left-hand column shows the relaxation time distributions obtained by finite element simulation (solid blue lines) and analytic solution (red crosses). The pore lengths $\rho T_\mu \equiv U$ and ρT_m predicted by analytic calculation (see Table I) are indicated by solid black lines above each distribution, and the constant macropore length L_m is indicated by a dotted black line above each distribution. The right-hand column shows the corresponding cumulative integrals from the finite element simulations (solid blue lines) and analytic solutions (dashed red lines), where $\hat{F}(T) = \int_{T'=0}^T \hat{f}(T') d(\log_{10} T')$. In each plot, the abscissa is scaled to effective pore size and the ordinate is scaled to total porosity.

change in the relaxation behavior in the macropore as α increases above unity, and the magnetization decay is no longer governed by the “fast diffusion” regime. The “intermediate” ($\alpha \sim 1$) and “slow” ($\alpha \gg 1$) diffusion regimes are characterized by spatially nonuniform magnetization. Both the analytic solution and simulations provide $m(\mathbf{r}, t)$ as a direct output, and the spatial variation of magnetization is visualized in Fig. 5. The spatial distribution of magnetization is shown across the 1D geometry with the macropore on the left ($x < L_m$) and the porous grain on the right ($x \geq L_m$) at various

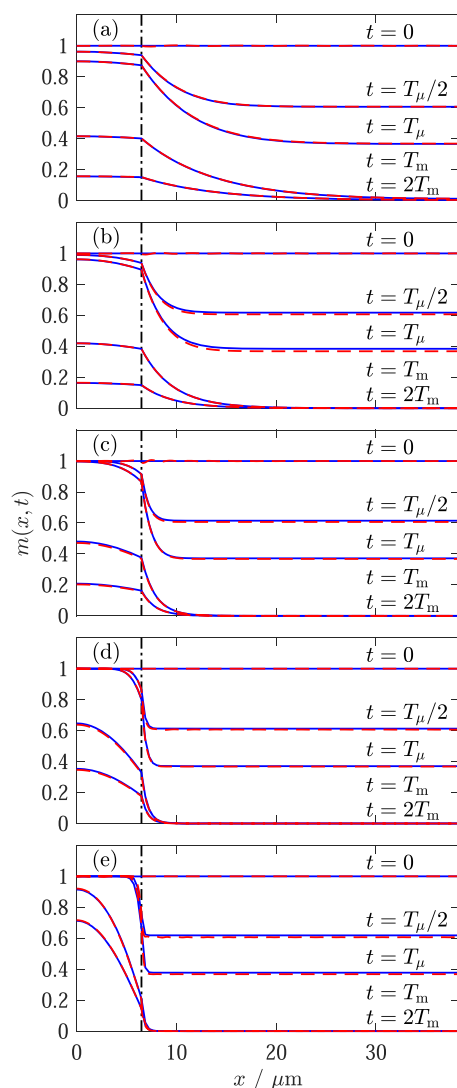


FIG. 5. Spatial variation of magnetization in a 1D μ GC analog geometry determined by finite element simulation (solid blue lines) and analytic solution (dashed red lines) at various times. Cases are shown for (a) $\rho = 1.5 \mu\text{m s}^{-1}$, (b) $7.5 \mu\text{m s}^{-1}$, (c) $37.5 \mu\text{m s}^{-1}$, (d) $187.5 \mu\text{m s}^{-1}$, and (e) $937.5 \mu\text{m s}^{-1}$. See text for other simulation parameters. In each plot, the vertical dotted-dashed black line indicates the boundary at L_m . Results from the simulations and the analytic solutions are indistinguishable in most cases.

multiples of T_μ and T_m ; note that these times vary with surface relaxivity, as listed in Table I. At the center of the grain, $x = W = 38.34 \mu\text{m}$, the magnetization reaches a plateau in all cases that corresponds to $\exp\{-t/T_\mu\}$. When the surface relaxation is weak [Fig. 5(a)], the influence of magnetization diffusing into the grain extends to $x \approx 25 \mu\text{m}$. However, at the other end of the scale when the surface relaxation is strong [Fig. 5(e)], magnetization diffusing into the grain relaxes immediately and the magnetization remains spatially uniform inside the grain. Conversely, when the surface relaxation is weak, then the magnetization in the macropore

remains approximately uniform, corresponding to the “fast diffusion” regime ($\alpha \approx 0.01$). However, when the surface relaxation is strong, we observe a dramatic spatial variation across the macropore, corresponding to relaxation in the “slow diffusion” regime ($\alpha \approx 6$). This visualization of the spatial variation in magnetization is extremely useful for interpreting the relaxation behavior observed in the distributions of Fig. 4. This spatial information is also available in 3D simulations, although it is less straightforward to inspect. A more general discussion of magnetization exchange

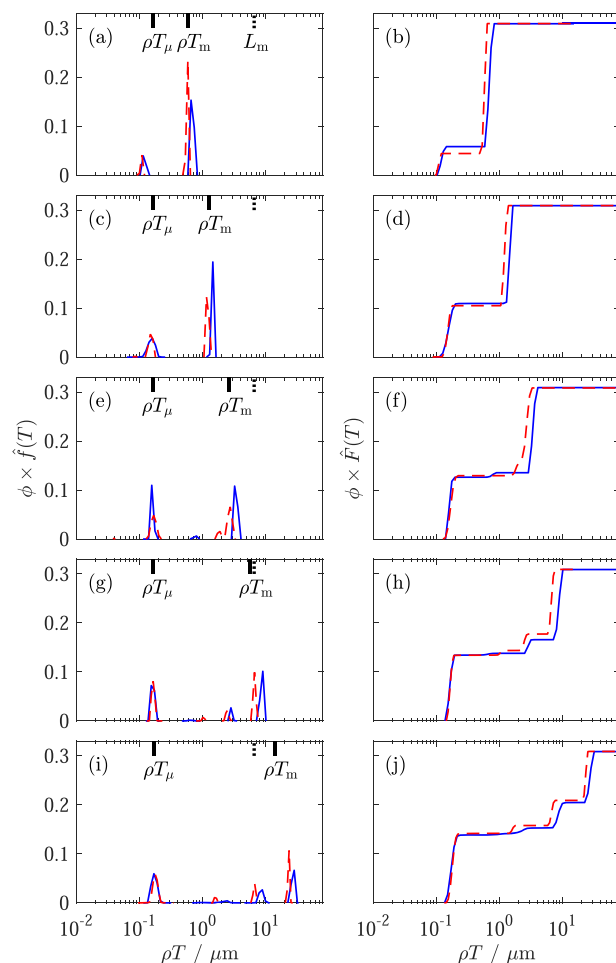


FIG. 6. Simulated relaxation behavior in the 3D μ GC geometry. The surface relaxivity varied as [(a) and (b)] $\rho = 1.5 \mu\text{m s}^{-1}$, [(c) and (d)] $7.5 \mu\text{m s}^{-1}$, [(e) and (f)] $37.5 \mu\text{m s}^{-1}$, [(g) and (h)] $187.5 \mu\text{m s}^{-1}$, and [(i) and (j)] $937.5 \mu\text{m s}^{-1}$. See text for other simulation parameters. The left-hand column shows the relaxation time distributions obtained by finite element simulation (solid blue lines) and random walk simulation (dashed red lines). The pore sizes $\rho T_\mu \equiv V_\mu/S_\mu$ and ρT_m estimated by finite element simulation are indicated by solid black lines above each distribution, and the constant macropore size $L_m \equiv V_m/S_m$, also determined by finite element simulation, is indicated by a dotted black line above each distribution. The right-hand column shows the corresponding cumulative integrals from the finite element simulations (solid blue lines) and random walk simulations (dashed red lines). In each plot, the abscissa is scaled to effective pore size and the ordinate is scaled to total porosity.

between domains has been presented previously by Arns and co-workers.^{48,49}

B. Comparison of finite element and random walk simulations in 3D

The relaxation time distributions generated from the 3D μ GC geometry by finite element and random walk simulations are compared in Fig. 6. The geometry was constructed with $A_m = 80 \mu\text{m}$, $2R_m/A_m = 1.23 = 2R_\mu/A_\mu$, and $A_\mu = 2 \mu\text{m}$. $T_b = \infty$ and $D_0 = 2.5 \times 10^{-9} \text{m}^2 \text{s}^{-1}$. For the finite element simulations, $\kappa = 2.43$ and T_μ (with $U = \rho T_\mu = 0.16 \mu\text{m}$) was calculated directly using the isolated pore geometry in Fig. 3 for each surface relaxivity value. The dominant macropore relaxation time was also estimated using the isolated pore geometry in Fig. 3 with an apparent surface relaxivity according to Eq. (11). The estimated relaxation times and corresponding pore sizes are listed in Table II. Note that the apparent surface relaxivity ρ_{app} in the macropores is substantially larger than the true surface relaxivity, especially when ρ is small. When the surface relaxation was weak [Figs. 6(a) and 6(c)], a bi-modal distribution was obtained from each simulation. Again, the effective macropore size ρT_m is much smaller than the true pore length L_m due to coupling. As the surface relaxivity increased through Figs. 6(e), 6(g), and 6(i), additional peaks were visible corresponding to multiple relaxation eigenmodes in the macropore when $\alpha \geq 1$. As in the 1D simulation, the fastest relaxing component is comparable to T_μ and this component is identical in the two simulations. It is interesting to note that the shortest relaxation time is slightly less than T_μ when the surface relaxation is weak [Fig. 6(a)]. This same phenomenon was present in the 1D simulations, albeit less obvious. The peaks at longer relaxation times in Fig. 6 differ slightly between the simulations. Consistently, the components attributed to relaxation in the macropore occur at shorter times in the random walk simulations than in the finite element simulations. The same temporal shift is visible in the cumulative distributions [Fig. 6 (right)], where it is clear that the amplitudes of the components are comparable. This discrepancy was attributed to the difference in detail of the macropore and grain interface between the two models. In the random walk simulations, the interface between macropore and grain is not well defined and can be considered a “rough” surface of finite depth with modified diffusion behavior. In the finite element model, this interface is well defined, is smooth, and has zero depth. Therefore, the grain surface area appears larger in the

random walk simulation than in the finite element simulation, leading to shorter macropore relaxation times. The finite element simulation could be modified to include a boundary layer of finite depth, or the boundary conditions adjusted, to replicate this effect. However, the discrepancy between the two sets of simulated data is considered small and we regard the results from the two models as consistent.

VI. VISUALIZING EXCHANGE

When inspecting the relaxation time distributions in Figs. 4(a) and 6(a), for example, one can make the naïve assignment of peaks in the bi-modal distributions to nuclear spins in the micropores (short relaxation time) and macropores (long relaxation time). However, this is incorrect as the ratio of the peak areas does not match the porosity fractions of the micropores and macropores, as demonstrated previously.³ Diffusion of magnetization between pores on the time scale of the measurement alters the position and amplitude of the observed components. To inspect the influence of this diffusive coupling, we can adjust the input parameters of our simulation and initialize the model with magnetization confined in the macropores or micropores. To visualize this exchange process, we use the 1D geometry of Fig. 2. The finite element simulation allows the initial spatial distribution of magnetization at $t = 0$ to be trivially modified. Such modifications could be achieved in the analytic solutions or random walk simulations but with significantly more effort. An example result is presented in Fig. 7 using the same input parameters as the simulations in Fig. 4(a) but with $m(x \leq L_m, 0) = 1$ (macropore domain) and $m(x > L_m, 0) = 0$ (grain domain) in Fig. 7(a), and $m(x \leq L_m, 0) = 0$ (macropore domain) and $m(x > L_m, 0) = 1$ (grain domain) in Fig. 7(c). When the magnetization is present initially in the macropore only [Fig. 7(a)], it diffuses a significant distance into the grain before relaxing. Despite this exchange process, the relaxation time distribution in Fig. 7(b) contains a single peak located close to T_m , corresponding to relaxation in the macropore. Despite extensive diffusion of magnetization into the grain, the microporous region acts only as a relaxing surface under conditions of “fast diffusion,” so a single relaxation time is observed. When the magnetization is present initially in the microporous grain only [Fig. 7(c)], it diffuses out into the macropore. The magnetization in the macropore increases at short time ($t \rightarrow T_\mu$), before decreasing again at long times ($t \rightarrow T_m$). Now, the resulting relaxation time distribution in Fig. 7(d) contains two peaks. The majority of the signal reflects the relaxation inside the grain, and the main peak is located near T_μ . However, magnetization that has diffused into the macropore relaxes at the slower rate and a notable peak is present near T_m . To confirm these observations, the sum of the data in Figs. 7(a) and 7(c) is shown in Fig. 7(e), where the spatial variation in magnetization is indistinguishable from the result in Fig. 5(a). Similarly, the relaxation time distributions in Figs. 7(b) and 7(d) were summed to give the distribution in Fig. 7(f). Also shown in Fig. 7(f) is the distribution repeated from Fig. 7(a). The peak positions are identical, and although the amplitudes appear different due to smoothing in the regularization step, the integral areas match. This manipulation of the system is a powerful tool for exploring the influence of exchange on measured relaxation time distributions.

Alternatively, the exchange of magnetization between environments is observed in multidimensional relaxation time correlations.

TABLE II. Variation in estimated micropore and macropore relaxation times as a function of surface relaxivity, and the corresponding pore size based on Eq. (3). Also included are the Brownstein-Tarr parameter $\alpha_m = \rho L_m/D_0$ for the macropore only and the grain-macropore coupling parameter C as defined in Eq. (13).

ρ ($\mu\text{m s}^{-1}$)	ρ_{app} ($\mu\text{m s}^{-1}$)	T_μ (ms)	ρT_μ (μm)	T_m (ms)	ρT_m (μm)	α_m	C
1.5	17.8	107.6	0.16	384.7	0.58	0.01	0.09
7.5	43.3	21.5	0.16	166.1	1.25	0.02	0.45
37.5	113.9	4.3	0.16	69.6	2.61	0.10	2.26
187.5	340.3	0.9	0.16	29.6	5.55	0.49	11.32
937.5	1179.2	0.2	0.16	14.7	13.78	2.44	56.58

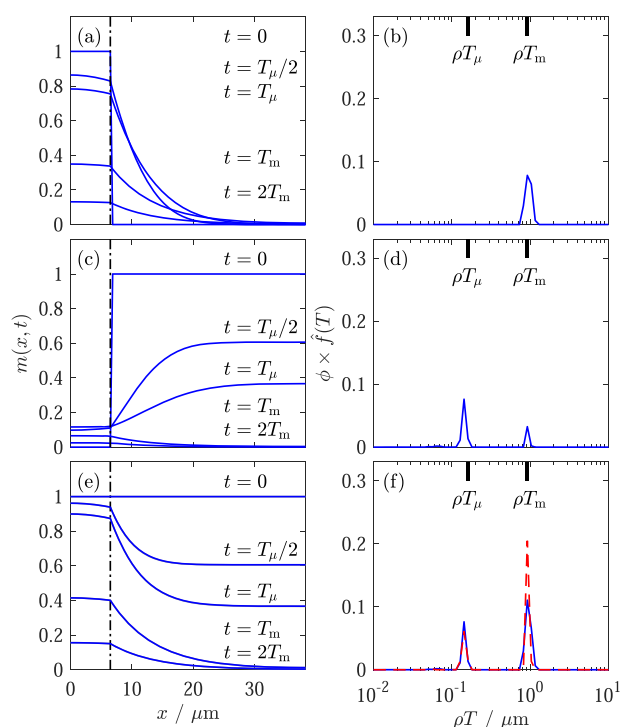


FIG. 7. Exploration of coupling in a 1D finite element simulation with $\rho = 1.5 \mu\text{m s}^{-1}$. The left-hand column shows the spatial variation of magnetization at various times. In each plot, the vertical dotted-dashed black line indicates the boundary at L_m . The initial distribution of magnetization was (a) $m(x < L_m, 0) = 1$, $m(x \geq L_m, 0) = 0$, and (c) $m(x < L_m, 0) = 0$, $m(x \geq L_m, 0) = 1$. The result in (e) was obtained by summing the data in (a) and (c) at each time increment and corresponds exactly to the initial condition of $m(x, t) = 1$. The right-hand column shows the corresponding relaxation time distributions obtained from the simulations (solid blue lines) with the abscissa scaled to effective pore size and the ordinate scaled to total porosity. The distribution in (f, solid blue line) was obtained by summing the distributions in (b) and (d). For comparison, the distribution from Fig. 4(a) is repeated in (f), dashed red line. The integral areas under the solid and dashed lines in (f) are equal. The pore lengths ρT_μ and ρT_m predicted by analytic calculation are indicated above each distribution.

T_1 - T_2 data were generated by finite element simulation in the 3D μGC model using the same parameters as in Fig. 6 with $\rho_2 \equiv \rho$. To observe so-called exchange peaks positioned away from the $T_1 = T_2$ diagonal,⁵⁰ it is necessary for $T_1 \neq T_2$,⁵⁰ so the simulations were performed with $\rho_1 = \rho_2/2$. The correlations plots are shown in Fig. 8 with the surface relaxivity increasing from (a) $\rho_2 = 1.5 \mu\text{m s}^{-1}$ to (e) $937.5 \mu\text{m s}^{-1}$. When the surface relaxation is weak [Fig. 8(a)], there is a significant exchange between the pore environments, as noted in the discussion of Fig. 7 above. Two main peaks are observed along the diagonal. Interestingly, these peaks exhibit a ratio $T_1/T_2 \approx 1.5$ rather than $T_1/T_2 = 2$ expected from the ratio of surface relaxivities. This result is attributed to spatial averaging of the magnetization reservoir over the macropore domain (given there is no bulk relaxation, $T_{1b} = T_{2b} = \infty$) that shifts the distribution toward the $T_1 = T_2$ diagonal. A significant off-diagonal exchange peak with positive amplitude is observed at $T_1 \approx 0.51 \text{ s}$ and $T_2 \approx 0.09 \text{ s}$. A much smaller exchange peak is also present with negative amplitude

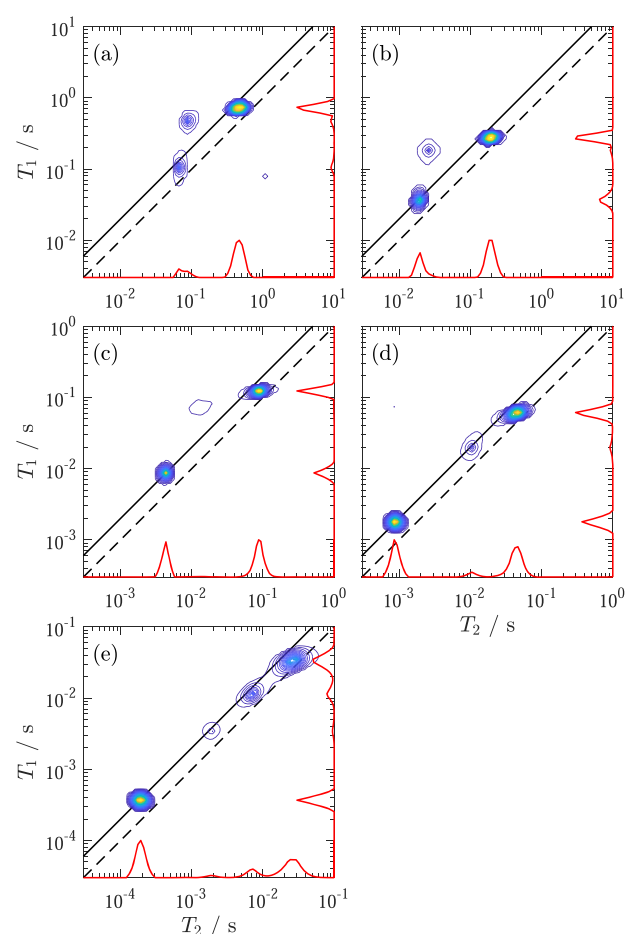


FIG. 8. T_1 - T_2 correlations generated by finite element simulation in the 3D μGC geometry. The transverse surface relaxivities varied as (a) $\rho_2 = 1.5 \mu\text{m s}^{-1}$, (b) $7.5 \mu\text{m s}^{-1}$, (c) $37.5 \mu\text{m s}^{-1}$, (d) $187.5 \mu\text{m s}^{-1}$, and (e) $937.5 \mu\text{m s}^{-1}$, and the ratio $\rho_2/\rho_1 = 2$. In each plot, the dashed diagonal line indicates $T_1 = T_2$ and the solid diagonal line indicates $T_1/T_2 = 2$. Peaks below the $T_1 = T_2$ diagonal have negative amplitude. Marginal T_1 and T_2 distributions (red lines) are included for clarity. See online version for references to color.

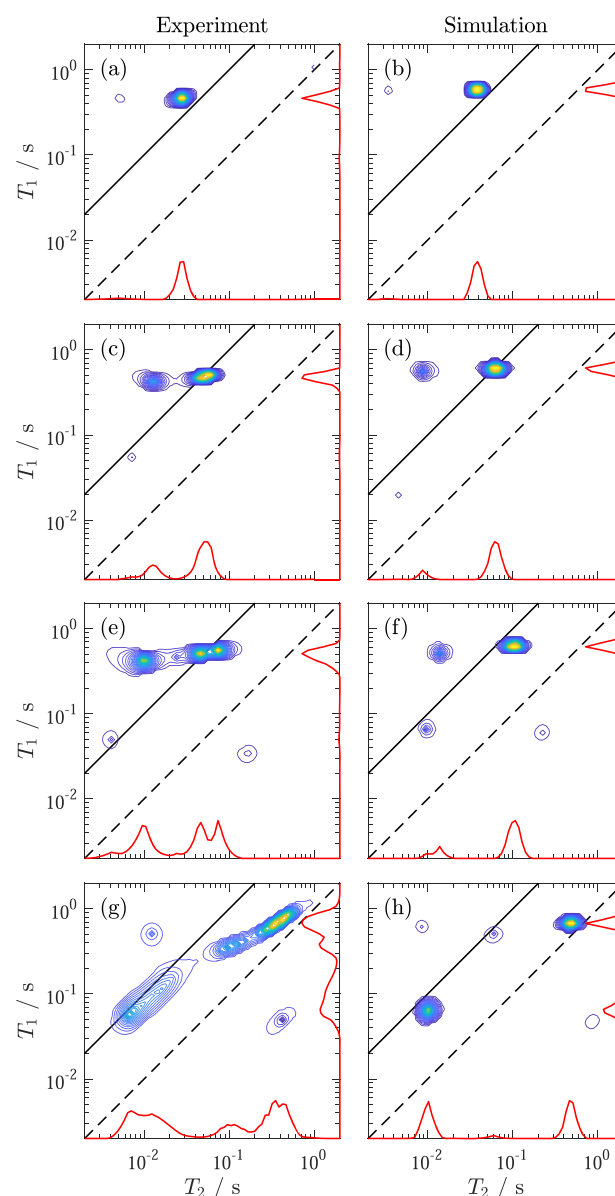
(corresponding to approximately 2% of the total signal) at $T_1 \approx 0.08 \text{ s}$ and $T_2 \approx 1.07 \text{ s}$ (the position of the peak is sensitive to the details of the inversion method²⁸). In general, components with $T_2 > T_1$ do not correspond to physical environments and this negative amplitude component is a consequence of magnetization transfer between the coupled longitudinal and transverse eigenmodes. Note that the marginal distributions are entirely positive. As the surface relaxivity increases through Figs. 8(b) and 8(c), the off-diagonal peaks become less pronounced and the negative amplitude component is no longer visible in the plots. When the surface relaxation is strong [Figs. 8(d) and 8(e)], the exchange peaks disappear entirely and multiple peaks are visible along the diagonal corresponding to "intermediate" and "slow" diffusion in the macropores. Figure 8(e) is a particularly interesting case. The peak at short relaxation times is associated with relaxation in the microporous grain and exhibits $T_1/T_2 = 2$ in agreement with the defined ratio $\rho_2/\rho_1 = 2$. The other

three peaks, associated with “slow diffusion” in the macropores, shift closer to the $T_1 = T_2$ diagonal at longer relaxation times, again due to the equal bulk relaxation times. Exchange of magnetization is occurring between the micropores and macropores, but the relaxation in the microporous grain is so extreme that the two environments appear uncoupled in the T_1 - T_2 correlation. Taken alone, the plot in Fig. 8(e) could be misinterpreted. The ability to examine the spatial magnetization profiles in simulation, as demonstrated in Figs. 5 and 7, elucidates the relaxation behavior observed in the correlation plots. We will use our finite element simulations to explore additional details of the T_1 - T_2 and T_2 - T_2 correlations in future work.

VII. SIMULATION AND EXPERIMENT

Finally, we demonstrate the utility of finite element methods by comparing experimental and simulated T_1 - T_2 correlations. The experimental results obtained on silica powders with a variety of grain sizes are shown in Fig. 9 (left). When the grains were relatively small, $R_m = 3$ – $17.5\ \mu\text{m}$ in Fig. 9(a), rapid exchange between the grain and macropores occurred on the time scale of the experiments and a single dominant peak was observed. As the grain size increased, through Figs. 9(c), 9(e), and 9(g), off-diagonal exchange peaks were observed consistent with diffusive coupling between two environments. In Fig. 9(e), two peaks are observed at long times near the $T_1/T_2 = 10$ diagonal and these multiple components are associated with the broad distribution of grain sizes ($R_m = 30$ – $100\ \mu\text{m}$) in this sample. It is interesting to observe that in Figs. 9(a), 9(c), and 9(e), multiple peaks are visible only in the T_2 marginal distribution and the T_1 marginal distribution is dominated by a single relaxation time component. This bias is attributed to details of the exchange process, where the off-diagonal peak at $T_1 \gg T_2$ is expected to have a larger amplitude than the off-diagonal peak at $T_2 \ll T_1$. The relatively low amplitude of the peaks at short relaxation times visible in Figs. 9(c) and 9(e) is attributed to weak surface relaxation in the microporous grain. The distribution in Fig. 9(g) is also interesting for its complexity. The off-diagonal exchange peaks are well defined and have significant and similar magnitude (albeit opposite sign). A broad, multimodal, and almost continuous distribution stretches diagonally across the plot, extending over two orders of magnitude in T_2 . This relaxation behavior is attributed to the range of grain sizes combined with “intermediate” diffusion in the macropores. The distribution shifts to a lower T_1/T_2 ratio at the longest relaxation times due to the influence of the bulk liquid relaxation, estimated as $T_{1b} \approx T_{2b} = 2\ \text{s}$ for de-ionized water measured on a low-field NMR instrument. Overall, the relatively long relaxation times combined with small micropore size suggest weak surface relaxation, and a ratio of $\rho_2/\rho_1 \sim 10$. Such a high ratio of surface relaxivities is uncommon in coupled porous media such as reservoir rocks⁵¹ where surface relaxation is dominated by paramagnetic moieties.⁵² However, in the absence of strong relaxation sinks, the surface relaxation is controlled by modified molecular dynamics⁵³ and high ρ_2/ρ_1 ratios have been observed previously on clean oxide surfaces, such as catalyst supports.⁵⁴

Relaxation of water in the powdered silicas was simulated using the 3D finite element μGC model. The geometry of each unit cell was constructed using the manufacturer’s quoted micropore size of $U = 4\ \text{nm}$ or $6\ \text{nm}$ as appropriate. The grain size was chosen



Here they are using R_g instead of the R_m used in the text

FIG. 9. Comparison of T_1 - T_2 correlations measured experimentally on water-saturated porous silica powders [(a), (c), (e), and (g)] and finite element simulations with matched physical parameters [(b), (d), (f), (h)]. The macroscopic grain sizes were (b) $R_g = 3\ \mu\text{m}$, (d) $20\ \mu\text{m}$, (f) $30\ \mu\text{m}$, and (h) $100\ \mu\text{m}$. In each plot, the dashed diagonal line indicates $T_1 = T_2$ and the solid diagonal line indicates $T_1/T_2 = 10$. Peaks below the $T_1 = T_2$ diagonal have negative amplitude. Marginal T_1 and T_2 distributions (red lines) are included for clarity. See online version for references to color.

based on the lower limit of the quoted ranges, so $R_m = 3, 20, 30$, or $100\ \mu\text{m}$, with $2R_m/A_m = 1$. The bulk diffusion coefficient was set at $D_0 = 2.3 \times 10^{-9}\ \text{m}^2\ \text{s}^{-1}$ for water at $25\ ^\circ\text{C}$, and the bulk relaxation times were $T_{1b} = T_{2b} = 2\ \text{s}$. The grain properties were set at $\phi_\mu = 0.4$ (estimated from experimental data) and $\kappa = 2.43$. Based on the experimental results, the surface relaxivities were set at ρ_1

$= 0.04 \mu\text{m s}^{-1}$ and $\rho_2 = 0.4 \mu\text{m s}^{-1}$. The simulations reflect a simplified version of the real samples: the range of grain sizes was not considered, and the simple cubic grain packing was retained rather than a more realistic structure such as a body centered cubic packing. Notwithstanding this simplification, the salient features of the experimental T_1 - T_2 correlations are captured in the corresponding simulated data in Fig. 9 (right). The exact position of the peaks is slightly different between the experiment and simulation, most likely due to subtle variations in surface relaxivity (surface chemistry) and porosity between samples. The same exchange phenomena are observed in the experiment and simulation, suggesting that the physical parameters (grain and pore size, porosity, and surface relaxivity) selected for the simulations are a close reflection of the sample properties. The distribution in Fig. 9(h) can be interpreted in terms of two exchange process: between the macropores and the grain, and within the macropores due to relaxation occurring outside the “fast diffusion” regime. In the latter case, the exchange process reflects the transfer of magnetization between longitudinal and transverse eigenmodes rather than diffusion of nuclear spins between different physical environments. This relaxation behavior accounts for some of the complexity in the experimental result in Fig. 9(g). Overall, the simulations provide a route to estimating the surface relaxivity parameters that are challenging to determine in coupled pore systems.

VIII. CONCLUSIONS

In this paper, we introduced finite element models of NMR measurements in coupled pore systems. For simplicity, we utilized the COMSOL Multiphysics software to solve the Bloch-Torrey equations for diffusing magnetization. The models were designed to reflect granular porous media consisting of spherical microporous grains separated by macroscopic interstices (macropores). Such materials include carbonate rocks, cements, catalyst supports, and biological cells. To validate these models, we based our simulations on an existing microporous grain consolidation (μGC) geometry.²⁹ In order to successfully mesh the geometries, it was necessary to represent the microporous grains as an effective medium with modified diffusion and relaxation properties. The finite element simulations in an analog 1D geometry were shown to be equivalent to analytic solutions, and the simulations in the 3D μGC model were comparable to random walk simulations. The finite element simulations did not capture the surface roughness of the grains present in the random walk simulations due to the use of a smooth interface between the macropore and microporous grain, but this detail could be represented by adjusted interface boundary conditions. Nevertheless, the salient features of (a) diffusive coupling and (b) Brownstein-Tarr “slow diffusion” in the macropores when the surface relaxivity was strong were reproduced correctly in the finite element simulations. Finally, we used the 3D simulations to replicate experimental results of water-saturated silica powder and hence estimate the longitudinal and transverse surface relaxivity parameters that are otherwise challenging to determine in the presence of diffusive coupling between relaxation environments.

The finite element simulations offer several advantages over existing forward modeling techniques to improve the interpretation of NMR relaxation time measurements in coupled porous media. Arbitrary grain structures can be constructed, or formed from $\mu\text{-CT}$

images of real materials, whereas the existing analytic solutions are limited to simple geometric forms. The finite element simulations solve the spatial distribution of magnetization directly, allowing convenient visualization of the time-dependent diffusion processes, even in 3D geometries. Modifications of the input finite element simulation parameters are trivial, such as the initial distribution of magnetization, allowing exchange processes to be explored. These simulations are also readily extensible to multidimensional NMR experiments, such as T_1 - T_2 or T_2 - T_2 correlations.

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REFERENCES

- ¹S. Davies and K. J. Packer, *J. Appl. Phys.* **67**, 3163 (1990).
- ²K. Brownstein and C. Tarr, *Phys. Rev. A* **19**, 2446 (1979).
- ³T. Ramakrishnan, L. Schwartz, E. Fordham, W. Kenyon, and D. Wilkinson, *Log Anal.* **40**, 260 (1999).
- ⁴T. Ramakrishnan, E. Fordham, L. Venkataramanan, M. Flaum, and L. Schwartz, in paper 1999-MMM presented at the SPWLA 40th Annual Logging Symposium, May 30–June 3, 1999.
- ⁵M. Fleury and J. Soualem, *J. Colloid Interface Sci.* **336**, 250 (2009).
- ⁶M. Hollewand and L. Gladden, *Chem. Eng. Res. Des.* **70**, 183 (1992).
- ⁷L. Baker, M. Renshaw, M. Mantle, A. Sederman, A. Wain, and L. Gladden, *Appl. Catal., A* **557**, 125 (2018).
- ⁸P. J. McDonald, J. P. Korb, J. Mitchell, and L. Monteilhet, *Phys. Rev. E* **72**, 011409 (2005).
- ⁹L. Monteilhet, J. P. Korb, J. Mitchell, and P. J. McDonald, *Phys. Rev. E* **74**, 061404 (2006).
- ¹⁰K. Song, J. Mitchell, H. Jaffel, and L. Gladden, *J. Phys. D: Appl. Phys.* **45**, 105302 (2012).
- ¹¹Y. Q. Song, L. Venkataramanan, M. D. Hürlimann, M. Flaum, P. Frulla, and C. Straley, *J. Magn. Reson.* **154**, 261 (2002).
- ¹²A. E. English, K. P. Whittall, M. L. G. Joy, and R. M. Henkelman, *Magn. Reson. Med.* **22**, 425 (1991).
- ¹³H. McConnell, *J. Chem. Phys.* **28**, 430 (1958).
- ¹⁴J. H. Lee, C. Labadie, C. S. Springer, and G. S. Harbison, *J. Am. Chem. Soc.* **115**, 7761 (1993).
- ¹⁵Y. Song, G. Carneiro, L. Schwartz, and D. Johnson, *Phys. Rev. Lett.* **113**, 235503 (2014).
- ¹⁶K. Washburn and P. Callaghan, *Phys. Rev. Lett.* **97**, 175502 (2006).
- ¹⁷M. van Landeghem, A. Haber, J. B. d’Espinose de Lacaillerie, and B. Blümich, *Concepts Magn. Reson., Part A* **36A**, 153 (2010).
- ¹⁸D. Bytchenkoff and S. Rodts, *J. Magn. Reson.* **208**, 4 (2011).
- ¹⁹V. Anand and G. Hirasaki, *Petrophysics* **48**, 289 (2007).
- ²⁰Z. Zhang, D. Johnson, and L. Schwartz, *Phys. Rev. E* **84**, 031129 (2011).
- ²¹L. Schwartz, D. Johnson, J. Mitchell, T. Chandrasekera, and E. Fordham, *Phys. Rev. E* **88**, 032813 (2013).
- ²²D. Johnson and L. Schwartz, *Phys. Rev. E* **90**, 032407 (2014).
- ²³G. Carneiro, A. Souza, A. Boyd, L. Schwartz, Y. Song, R. Azeredo, W. Trevizan, B. Santos, E. Rios, and V. Machado, in paper 2014-AA presented at the SPWLA 55th Annual Logging Symposium, May 18–22, 2014.

- ²⁴D. Johnson and L. Schwartz, *Phys. Rev. E* **91**, 062406 (2015).
- ²⁵C. Arns, Y. Melean, L. Burcaw, P. Callaghan, and K. Washburn, in paper SCA2009-44 presented at the Society of Core Analysts Annual Symposium Held in Noordwijk, The Netherlands, September 27–30, 2009.
- ²⁶C. Arns, T. AlGhamdi, and J. Arns, *New J. Phys.* **13**, 015004 (2010).
- ²⁷E. Fordham and J. Mitchell, *Microporous Mesoporous Mater.* **269**, 35 (2018).
- ²⁸T. Chandrasekera and J. Mitchell, *Phys. Rev. E* **98**, 043308 (2018).
- ²⁹L. Schwartz, in paper SPWLA-1990-Y presented at the SPWLA 31st Annual Logging Symposium, June 24–27, 1990.
- ³⁰H. Torrey, *Phys. Rev.* **104**, 563 (1956).
- ³¹P. P. Mitra, P. N. Sen, and L. M. Schwartz, *Phys. Rev. B* **47**, 8565 (1993).
- ³²L. Latour, P. Mitra, R. Kleinberg, and C. Sotak, *J. Magn. Reson., Ser. A* **101**, 342 (1993).
- ³³P. Sen, L. Schwartz, P. Mitra, and B. Halperin, *Phys. Rev. B* **49**, 215 (1994).
- ³⁴G. Archie, *Trans. AIME* **146**, 54 (1942).
- ³⁵C. Davies, J. Griffith, A. Sederman, L. Gladden, and M. Johns, *J. Magn. Reson.* **187**, 170 (2007).
- ³⁶S. Torquato and I. Kim, *Appl. Phys. Lett.* **55**, 1847 (1989).
- ³⁷M. Leibig, *J. Phys. A: Math. Gen.* **26**, 3349 (1993).
- ³⁸D. Bergman, K. Dunn, L. Schwartz, and P. Mitra, *Phys. Rev. E* **51**, 3393 (1995).
- ³⁹E. Fordham, L. Venkataramanan, J. Mitchell, and A. Valori, *Diff. Fund.* **29**, 2 (2017); available at [https://diffusion.uni-leipzig.de/pdf/volume29/diff_fund_29\(2017\)2.pdf](https://diffusion.uni-leipzig.de/pdf/volume29/diff_fund_29(2017)2.pdf)
- ⁴⁰J. D. Wilson, *J. Mater. Sci.* **27**, 3911 (1992).
- ⁴¹H. Y. Carr and E. M. Purcell, *Phys. Rev.* **94**, 630 (1954).
- ⁴²S. Meiboom and D. Gill, *Rev. Sci. Instrum.* **29**, 688 (1958).
- ⁴³T. Chandrasekera, J. Mitchell, E. Fordham, L. Gladden, and M. Johns, *J. Magn. Reson.* **194**, 156 (2008).
- ⁴⁴R. L. Vold, J. S. Waugh, M. P. Klein, and D. E. Phelps, *J. Chem. Phys.* **48**, 3831 (1968).
- ⁴⁵L. Venkataramanan, Y. Q. Song, and M. D. Hürlimann, *IEEE Trans. Signal Process.* **50**, 1017 (2002).
- ⁴⁶J. Mitchell, T. Chandrasekera, and L. Gladden, *Prog. Nucl. Magn. Reson. Spectrosc.* **62**, 34 (2012).
- ⁴⁷R. Kleinberg, *Exp. Methods Phys. Sci.* **35**, 337 (1999).
- ⁴⁸N. Alhwerty, I. Shikhov, J. Arns, and C. Arns, in paper YYY presented at the SPWLA 57th Annual Logging Symposium Held in Reykjavik, Iceland, June 25–29, 2016.
- ⁴⁹Y. Zheng, I. Shikhov, L. Gbayan, J. Arns, and C. Arns, *J. Contam. Hydrol.* **212**, 143 (2018).
- ⁵⁰Y. Song, L. Zielinski, and S. Ryu, *Phys. Rev. Lett.* **100**, 248002 (2008).
- ⁵¹R. Kleinberg, S. Farooqui, and M. Horsfield, *J. Colloid Interface Sci.* **158**, 195 (1993).
- ⁵²S. Godefroy, J. Korb, M. Fleury, and R. Bryant, *Phys. Rev. E* **64**, 021605 (2001).
- ⁵³G. Liu, Y. Li, and J. Jonas, *J. Chem. Phys.* **95**, 6892 (1991).
- ⁵⁴D. Weber, J. Mitchell, J. McGregor, and L. F. Gladden, *J. Phys. Chem. C* **113**, 6610 (2009).